Nashira Nasrudeen

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Profile

Software Engineer and **MEng graduate** in Artificial Intelligence with 2+ years of production software experience and research experience in Responsible AI, LLM evaluation, and uncertainty quantification.

Education

Toronto Metropolitan University — Toronto, ON
Master of Engineering in Artificial Intelligence, GPA: 4.17/4.33 — *Sep. 2025 – Aug. 2026*

University of Moratuwa — Sri Lanka
Bachelor of Science (Honours) in Materials Science & Engineering — *2017 – 2022*

Research Experience

Graduate Student Researcher | *Trustworthy AI Lab, Toronto Metropolitan University* — Mar. 2026 – Aug. 2026

- Developed a Python evaluation framework for uncertainty quantification in LLM-as-a-Judge toxicity detection, evaluating 14+ semantic, NLI, lexical, and consistency-based methods across multiple language models and safety datasets.
- Evaluated Adaptive Conformal Semantic Entropy (ACSE), a semantic UQ method, for uncertainty-aware toxicity classification and compared its ranking and calibration behavior with alternative UQ approaches, with NLI and lexical methods achieving AUROC up to 0.817 and 0.895 on ToxicChat.
- Investigated failure modes in consistency-based UQ methods for LLM-as-a-Judge toxicity detection, including single-cluster collapse and soft-assignment compression, identifying cases where these methods failed to detect confidently incorrect model judgments.

Professional Experience

Software Engineer — Aug. 2023 – Sep. 2025
Encyte (Pvt) Ltd — *Sri Lanka*

- Built backend microservices and REST APIs (Node.js, Express, TypeScript, MongoDB) for a multi-tenant maintenance-management SaaS serving 3 enterprise clients across 7+ services (work orders, assets, customers, sites, analytics, API gateway).
- Developed React/Next.js frontends from Figma designs and delivered a contract-management module end to end, including clause and template versioning, staged review-and-approval workflows, pricing, reporting, and email notifications.
- Implemented role-based access control with Auth0 and JWT (page-level restrictions, per-role data scoping); improved API performance by 30% through query and schema refactoring, standardized error handling across services, and load-tested with Artillery.
- Led migration of core services to NestJS (tenant and user services) and mentored 4+ interns and the web team through JavaScript, React/Redux, and backend training sessions.

Projects

Emotion-Conditioned Conversational Language Modeling — GitHub

- Fine-tuned BART on the DailyDialog dataset for emotion-conditioned response generation, improving 3-class emotion classification accuracy from 50.95% to 54.70% and macro-F1 from 37.12% to 40.25%.
- Evaluated conversational generation performance across emotion-conditioned settings and analyzed the effect of incorporating emotional context into language modeling.

AI Web-Form Agent — GitHub

- Built a TypeScript service integrating Gemini with Playwright browser automation through tool-based actions and an iterative observe-and-act workflow.
- Implemented an HTTP API, input validation, retry handling, scheduled execution, and job-overlap protection for reliable automated runs.

Adversarial Robustness in Differentially Private Federated Learning — GitHub

- Built a differentially private federated learning pipeline for a Human Activity Recognition model, using a 1D CNN, DP-SGD, gradient clipping, and Gaussian noise, across simulated clients, to evaluate adversarial robustness under white-box PGD attacks.
- Implemented Federated Adversarial Training to improve robustness under privacy constraints, increasing robust accuracy from 9.16% to 29.59% at adversarial $\epsilon = 0.05$ and analyzing privacy-utility-robustness trade-offs.

Technical Skills

AI & ML: PyTorch, Scikit-learn, NumPy, pandas, NLP
Languages: Python, TypeScript, JavaScript, SQL, C++
Full-stack: FastAPI, Node.js, Express, REST APIs, React, Next.js, Redux, MongoDB, PostgreSQL
Tools: Git, Docker, Linux, AWS, GCP, Playwright